



MATH130: Quadratic Functions

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Learning Objectives

- Solve quadratic equations by factoring and the quadratic formula
- Find the vertex and axis of symmetry of a parabola
- Solve absolute value equations
- Interpret the discriminant to determine the nature of roots

Simplify each expression completely. Show all steps and circle your final answer.

Absolute value equations

1. Solve: $|1x + -6| = 6$.

$$|1x + -6| = 6$$

Answer: _____

2. Solve: $|1x + 0| = 2$.

$$|1x + 0| = 2$$

Answer: _____

3. Solve: $|4x + -2| = 7$.

$$|4x + -2| = 7$$

Answer: _____

4. Solve: $|2x + 1| = 10$.

$$|2x + 1| = 10$$

Answer: _____

5. Solve: $|2x + -3| = 12$.

$$|2x + -3| = 12$$

Answer: _____

6. Solve: $|2x + -5| = 5$.

$$|2x + -5| = 5$$

Answer: _____

7. Solve: $|1x + -2| = 8$.

$$|1x + -2| = 8$$

Answer: _____

8. Solve: $|1x + -2| = 9$.

$$|1x + -2| = 9$$

Answer: _____

Solving quadratic equations

9. Solve: $x^2 + 3x + -18 = 0$ using the quadratic formula.

$$x^2 + 3x - 18 = 0$$

Answer: _____

10. Solve by factoring: $1x^2 + -4x + 3 = 0$.

$$x^2 - 4x + 3 = 0$$

Answer: _____

11. Solve: $x^2 + 3x + -4 = 0$ using the quadratic formula.

$$x^2 + 3x - 4 = 0$$

Answer: _____

12. Solve by factoring: $1x^2 + -3x + 0 = 0$.

$$x^2 - 3x = 0$$

Answer: _____

13. Solve: $x^2 + 0x + -25 = 0$ using the quadratic formula.

$$x^2 - 25 = 0$$

Answer: _____

14. Solve by factoring: $2x^2 + 12x + 10 = 0$.

$$2x^2 + 12x + 10 = 0$$

Answer: _____

15. Solve: $x^2 + 4x + -12 = 0$ using the quadratic formula.

$$x^2 + 4x - 12 = 0$$

Answer: _____

16. Solve by factoring: $2x^2 + 4x + -30 = 0$.

$$2x^2 + 4x - 30 = 0$$

Answer: _____

17. Solve: $x^2 + 4x + -5 = 0$ using the quadratic formula.

$$x^2 + 4x - 5 = 0$$

Answer: _____

18. Solve by factoring: $1x^2 + 2x + 1 = 0$.

$$x^2 + 2x + 1 = 0$$

Answer: _____

19. Solve: $x^2 + -1x + -12 = 0$ using the quadratic formula.

$$x^2 - 1x - 12 = 0$$

Answer: _____

20. Solve by factoring: $2x^2 + 4x + -16 = 0$.

$$2x^2 + 4x - 16 = 0$$

Answer: _____

21. Solve: $x^2 + 5x + -6 = 0$ using the quadratic formula.

$$x^2 + 5x - 6 = 0$$

Answer: _____

22. Solve by factoring: $1x^2 + -5x + 6 = 0$.

$$x^2 - 5x + 6 = 0$$

Answer: _____

23. Solve: $x^2 + -1x + -20 = 0$ using the quadratic formula.

$$x^2 - 1x - 20 = 0$$

Answer: _____

Vertex of a parabola

24. Find the vertex of the parabola $f(x) = -2x^2 + 8x + -9$.

$$f(x) = -2x^2 + 8x - 9$$

Answer: _____

25. Find the vertex of the parabola $f(x) = 2x^2 + -4x + -4$.

$$f(x) = 2x^2 - 4x - 4$$

Answer: _____

26. Find the vertex of the parabola $f(x) = 2x^2 + -12x + 24$.

$$f(x) = 2x^2 - 12x + 24$$

Answer: _____

27. Find the vertex of the parabola $f(x) = 2x^2 + -16x + 23$.

$$f(x) = 2x^2 - 16x + 23$$

Answer: _____

28. Find the vertex of the parabola $f(x) = -1x^2 + 6x + -12$.

$$f(x) = -1x^2 + 6x - 12$$

Answer: _____

29. Find the vertex of the parabola $f(x) = -2x^2 + -4x + 5$.

$$f(x) = -2x^2 - 4x + 5$$

Answer: _____

30. Find the vertex of the parabola $f(x) = 1x^2 + 6x + 2$.

$$f(x) = 1x^2 + 6x + 2$$

Answer: _____



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ANSWER KEY & SOLUTIONS

Topics: Solving quadratic equations, Absolute value equations, Vertex of a parabola. All answers verified by independent computation.

Solutions

Absolute value equations

1. Solve: $|1x + -6| = 6$.

$$|1x + -6| = 6$$

→ Split into two cases: $1x + -6 = 6$ or $1x + -6 = -6$.

→ Case 1: $1x = 6 - -6 = 12 \Rightarrow x = 12/1 = 12$.

→ Case 2: $1x = -6 - -6 = 0 \Rightarrow x = 0/1 = 0$.

Answer: $x = 0$ or $x = 12$

2. Solve: $|1x + 0| = 2$.

$$|1x + 0| = 2$$

→ Split into two cases: $1x + 0 = 2$ or $1x + 0 = -2$.

→ Case 1: $1x = 2 - 0 = 2 \Rightarrow x = 2/1 = 2$.

→ Case 2: $1x = -2 - 0 = -2 \Rightarrow x = -2/1 = -2$.

Answer: $x = -2$ or $x = 2$

3. Solve: $|4x + -2| = 7$.

$$|4x + -2| = 7$$

→ Split into two cases: $4x + -2 = 7$ or $4x + -2 = -7$.

→ Case 1: $4x = 7 - -2 = 9 \Rightarrow x = 9/4 = 9/4$.

→ Case 2: $4x = -7 - -2 = -5 \Rightarrow x = -5/4 = -5/4$.

Answer: $x = -5/4$ or $x = 9/4$

4. Solve: $|2x + 1| = 10$.

$$|2x + 1| = 10$$

→ Split into two cases: $2x + 1 = 10$ or $2x + 1 = -10$.

→ Case 1: $2x = 10 - 1 = 9 \Rightarrow x = 9/2 = 9/2$.

→ Case 2: $2x = -10 - 1 = -11 \Rightarrow x = -11/2 = -11/2$.

Answer: $x = -11/2$ or $x = 9/2$

5. Solve: $|2x + -3| = 12$.

$$|2x + -3| = 12$$

→ Split into two cases: $2x + -3 = 12$ or $2x + -3 = -12$.

→ Case 1: $2x = 12 - -3 = 15 \Rightarrow x = 15/2 = 15/2$.

→ Case 2: $2x = -12 - -3 = -9 \Rightarrow x = -9/2 = -9/2$.

Answer: $x = -9/2$ or $x = 15/2$

6. Solve: $|2x + -5| = 5$.

$$|2x + -5| = 5$$

→ Split into two cases: $2x + -5 = 5$ or $2x + -5 = -5$.

→ Case 1: $2x = 5 - -5 = 10 \Rightarrow x = 10/2 = 5$.

→ Case 2: $2x = -5 - -5 = 0 \Rightarrow x = 0/2 = 0$.

Answer: $x = 0$ or $x = 5$

7. Solve: $|1x + -2| = 8$.

$$|1x + -2| = 8$$

→ Split into two cases: $1x + -2 = 8$ or $1x + -2 = -8$.

→ Case 1: $1x = 8 - -2 = 10 \Rightarrow x = 10/1 = 10$.

→ Case 2: $1x = -8 - -2 = -6 \Rightarrow x = -6/1 = -6$.

Answer: $x = -6$ or $x = 10$

8. Solve: $|1x + -2| = 9$.

$$|1x + -2| = 9$$

→ Split into two cases: $1x + -2 = 9$ or $1x + -2 = -9$.

→ Case 1: $1x = 9 - -2 = 11 \Rightarrow x = 11/1 = 11$.

→ Case 2: $1x = -9 - -2 = -7 \Rightarrow x = -7/1 = -7$.

Answer: $x = -7$ or $x = 11$

Solving quadratic equations

9. Solve: $x^2 + 3x - 18 = 0$ using the quadratic formula.

$$x^2 + 3x - 18 = 0$$

→ Discriminant: $b^2 - 4ac = (3)^2 - 4(1)(-18) = 81$.

→ $x = (-3 \pm \sqrt{81}) / 2$.

→ $x = -6$ or $x = 3$.

Answer: $x = -6$ or $x = 3$

10. Solve by factoring: $1x^2 + -4x + 3 = 0$.

$$x^2 - 4x + 3 = 0$$

→ Factor: $1(x - (1))(x - (3)) = 0$.

→ $x = 1$ or $x = 3$.

Answer: $x = 1$ or $x = 3$

11. Solve: $x^2 + 3x - 4 = 0$ using the quadratic formula.

$$x^2 + 3x - 4 = 0$$

→ Discriminant: $b^2 - 4ac = (3)^2 - 4(1)(-4) = 25$.

→ $x = (-3 \pm \sqrt{25}) / 2$.

→ $x = -4$ or $x = 1$.

Answer: $x = -4$ or $x = 1$

12. Solve by factoring: $1x^2 + -3x + 0 = 0$.

$$x^2 - 3x = 0$$

→ Factor: $1(x - (3))(x - (0)) = 0$.

→ $x = 0$ or $x = 3$.

Answer: $x = 0$ or $x = 3$

13. Solve: $x^2 + 0x - 25 = 0$ using the quadratic formula.

$$x^2 - 25 = 0$$

→ Discriminant: $b^2 - 4ac = (0)^2 - 4(1)(-25) = 100$.

→ $x = (-0 \pm \sqrt{100}) / 2$.

→ $x = -5$ or $x = 5$.

Answer: $x = -5$ or $x = 5$

14. Solve by factoring: $2x^2 + 12x + 10 = 0$.

$$2x^2 + 12x + 10 = 0$$

→ Factor: $2(x - (-1))(x - (-5)) = 0$.

→ $x = -5$ or $x = -1$.

Answer: $x = -5$ or $x = -1$

15. Solve: $x^2 + 4x - 12 = 0$ using the quadratic formula.

$$x^2 + 4x - 12 = 0$$

→ Discriminant: $b^2 - 4ac = (4)^2 - 4(1)(-12) = 64$.

→ $x = (-4 \pm \sqrt{64}) / 2$.

→ $x = -6$ or $x = 2$.

Answer: $x = -6$ or $x = 2$

16. Solve by factoring: $2x^2 + 4x - 30 = 0$.

$$2x^2 + 4x - 30 = 0$$

→ Factor: $2(x - (3))(x - (-5)) = 0$.

→ $x = -5$ or $x = 3$.

Answer: $x = -5$ or $x = 3$

17. Solve: $x^2 + 4x - 5 = 0$ using the quadratic formula.

$$x^2 + 4x - 5 = 0$$

→ Discriminant: $b^2 - 4ac = (4)^2 - 4(1)(-5) = 36$.

→ $x = (-4 \pm \sqrt{36}) / 2$.

→ $x = -5$ or $x = 1$.

Answer: $x = -5$ or $x = 1$

18. Solve by factoring: $1x^2 + 2x + 1 = 0$.

$$x^2 + 2x + 1 = 0$$

→ Factor: $1(x - (-1))(x - (-1)) = 0$.

→ $x = -1$ or $x = -1$.

Answer: $x = -1$ or $x = -1$

19. Solve: $x^2 + -1x + -12 = 0$ using the quadratic formula.

$$x^2 - 1x - 12 = 0$$

→ Discriminant: $b^2 - 4ac = (-1)^2 - 4(1)(-12) = 49$.

→ $x = (1 \pm \sqrt{49}) / 2$.

→ $x = -3$ or $x = 4$.

Answer: $x = -3$ or $x = 4$

20. Solve by factoring: $2x^2 + 4x - 16 = 0$.

$$2x^2 + 4x - 16 = 0$$

→ Factor: $2(x - (2))(x - (-4)) = 0$.

→ $x = -4$ or $x = 2$.

Answer: $x = -4$ or $x = 2$

21. Solve: $x^2 + 5x - 6 = 0$ using the quadratic formula.

$$x^2 + 5x - 6 = 0$$

→ Discriminant: $b^2 - 4ac = (5)^2 - 4(1)(-6) = 49$.

→ $x = (-5 \pm \sqrt{49}) / 2$.

→ $x = -6$ or $x = 1$.

Answer: $x = -6$ or $x = 1$

22. Solve by factoring: $1x^2 + -5x + 6 = 0$.

$$x^2 - 5x + 6 = 0$$

→ Factor: $1(x - (2))(x - (3)) = 0$.

→ $x = 2$ or $x = 3$.

Answer: $x = 2$ or $x = 3$

23. Solve: $x^2 + -1x + -20 = 0$ using the quadratic formula.

$$x^2 - 1x - 20 = 0$$

→ Discriminant: $b^2 - 4ac = (-1)^2 - 4(1)(-20) = 81$.

→ $x = (-(-1) \pm \sqrt{81}) / 2$.

→ $x = -4$ or $x = 5$.

Answer: $x = -4$ or $x = 5$

Vertex of a parabola

24. Find the vertex of the parabola $f(x) = -2x^2 + 8x + -9$.

$$f(x) = -2x^2 + 8x - 9$$

$$\rightarrow h = -b / (2a) = -(8) / (2*-2) = 2.$$

$$\rightarrow k = f(2) = -1.$$

$$\rightarrow \text{Vertex: } (2, -1).$$

Answer: Vertex = (2, -1)

25. Find the vertex of the parabola $f(x) = 2x^2 + -4x + -4$.

$$f(x) = 2x^2 - 4x - 4$$

$$\rightarrow h = -b / (2a) = -(-4) / (2*2) = 1.$$

$$\rightarrow k = f(1) = -6.$$

$$\rightarrow \text{Vertex: } (1, -6).$$

Answer: Vertex = (1, -6)

26. Find the vertex of the parabola $f(x) = 2x^2 + -12x + 24$.

$$f(x) = 2x^2 - 12x + 24$$

$$\rightarrow h = -b / (2a) = -(-12) / (2*2) = 3.$$

$$\rightarrow k = f(3) = 6.$$

$$\rightarrow \text{Vertex: } (3, 6).$$

Answer: Vertex = (3, 6)

27. Find the vertex of the parabola $f(x) = 2x^2 + -16x + 23$.

$$f(x) = 2x^2 - 16x + 23$$

$$\rightarrow h = -b / (2a) = -(-16) / (2*2) = 4.$$

$$\rightarrow k = f(4) = -9.$$

$$\rightarrow \text{Vertex: } (4, -9).$$

Answer: Vertex = (4, -9)

28. Find the vertex of the parabola $f(x) = -1x^2 + 6x + -12$.

$$f(x) = -1x^2 + 6x - 12$$

$$\rightarrow h = -b / (2a) = -(6) / (2*-1) = 3.$$

$$\rightarrow k = f(3) = -3.$$

$$\rightarrow \text{Vertex: } (3, -3).$$

Answer: Vertex = (3, -3)

29. Find the vertex of the parabola $f(x) = -2x^2 + -4x + 5$.

$$f(x) = -2x^2 - 4x + 5$$

$$\rightarrow h = -b / (2a) = -(-4) / (2*-2) = -1.$$

$$\rightarrow k = f(-1) = 7.$$

$$\rightarrow \text{Vertex: } (-1, 7).$$

Answer: Vertex = $(-1, 7)$

30. Find the vertex of the parabola $f(x) = 1x^2 + 6x + 2$.

$$f(x) = 1x^2 + 6x + 2$$

$$\rightarrow h = -b / (2a) = -(6) / (2*1) = -3.$$

$$\rightarrow k = f(-3) = -7.$$

$$\rightarrow \text{Vertex: } (-3, -7).$$

Answer: Vertex = $(-3, -7)$
